

Eigenvalue Quick Reference

Concise revision notes for eigenvalue equations, checks, and basic matrix types.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Eigenvectors are directions that survive multiplication by A . The matrix only stretches, shrinks, reverses, or rotates through a scalar factor.

The defining equation is

$$Ax = \lambda x, \quad x \neq 0.$$

Move all terms to one side:

$$(A - \lambda I)x = 0.$$

For a nonzero solution, the matrix $A - \lambda I$ must be singular:

$$\det(A - \lambda I) = 0.$$

2 Characteristic Equation

For a 2×2 matrix, the characteristic equation can be written as

$$\lambda^2 - \text{trace}(A)\lambda + \det A = 0.$$

This is often the fastest way to check arithmetic.

Quantity	Eigenvalue check
Trace	Sum of eigenvalues.
Determinant	Product of eigenvalues.
Zero eigenvalue	Matrix is singular.

3 Basic Examples

Matrix type	Eigenvalue memory
Projection P with $P^2 = P$	Eigenvalues are 0 or 1.
Permutation swapping two coordinates	Eigenvalues often include 1 and -1 .
Plane rotation by 90°	Eigenvalues are complex: i and $-i$.

4 Growth and Stability

For repeated multiplication A^k :

$$|\lambda| < 1$$

means decay in that eigenvector direction.

$$|\lambda| > 1$$

means growth in that direction.

$$\lambda = 1$$

means that direction stays fixed.

5 Common Mistakes

Mistake	Correct exam habit
Allowing $x = 0$ as an eigenvector.	Eigenvectors must be nonzero.
Solving $Ax = 0$ for every eigenvector.	Solve $(A - \lambda I)x = 0$.
Forgetting trace and determinant checks.	Use them after solving for λ .
Assuming all rotations have real eigenvectors.	Plane rotations can have complex eigenvalues.

6 Final Exam Checklist

Checklist.

1. Write $Ax = \lambda x$.
2. Form $A - \lambda I$.
3. Solve $\det(A - \lambda I) = 0$.
4. Check trace and determinant.
5. For each λ , solve $(A - \lambda I)x = 0$.
6. Use $|\lambda|$ to judge growth or decay for powers.